

Appendix A — Pseudocode

The following pseudocode summarises the simulation engine (full runnable source: /sim/model.py in the project files, Deliverable #3).

```
BEGIN SmartHospitalNetworkSimulation
DEFINE zones[4] // Patient Monitoring, Medical Records, Lab & Pharmacy, Emergency
DEFINE topologies[4] = { STAR, BUS, MESH, HYBRID }
DEFINE protocols[6] = { TCP, UDP, HTTP, DNS, FTP, ICMP }
DEFINE scenarios[3] = { NORMAL, PEAK_MONITORING, DISRUPTION }

FOR EACH topology IN topologies:
  SET avg_hops, contention_factor, redundancy_paths, core_capacity
  // from topology characteristics table
  FOR EACH protocol IN protocols:
    SET packet_size, base_rate, active_flows, is_reliable, sensitivity
    FOR EACH scenario IN scenarios:
      SET load_mult, capacity_mult, extra_loss // scenario multipliers
      capacity <- core_capacity * capacity_mult
      offered_load <- base_rate * load_mult * active_flows
      // * packet_size * contention_factor
      utilization <- MIN(offered_load / capacity, 0.999)

      delays <- [] loss_values <- []
      REPEAT 200 TIMES: // Monte-Carlo trials
        u <- utilization + SMALL_RANDOM_NOISE
        queue_delay <- M_M_1_DELAY(u, capacity, packet_size)
        backoff_delay <- CONTENTION_BACKOFF(contention_factor, u)
        total_delay <- (queue_delay + PROC_DELAY + backoff_delay)
        // * avg_hops + PROP_DELAY * avg_hops
        total_delay <- total_delay * SENSITIVITY_JITTER(sensitivity)
        APPEND total_delay TO delays

        congestion_loss <- MAX(0, u - 0.60) * 1.1
        collision_loss <- 0.018 * contention_factor^1.6 * u
        scenario_loss <- extra_loss / SQRT(redundancy_paths)
        loss <- (congestion_loss + collision_loss) / redundancy_paths
        // + scenario_loss
        IF is_reliable: loss <- loss * 0.4 // retransmission recovery
        APPEND CLIP(loss, 0, 0.5) TO loss_values
      END REPEAT

      mean_delay <- MEAN(delays)
      jitter <- STDDEV(delays)
      mean_loss <- MEAN(loss_values)
      pdr <- (1 - mean_loss) * 100
      throughput <- offered_load * (1 - mean_loss)
      STORE (topology, protocol, scenario, mean_delay, jitter,
        mean_loss, pdr, throughput, utilization)
    END FOR (scenario)
  END FOR (protocol)
END FOR (topology)

EXPORT all_results TO results_full.csv
GROUP BY (topology, scenario) -> AVERAGE across protocols
EXPORT summary TO results_summary.csv
GENERATE comparison charts (delay, jitter, loss, PDR, throughput, util.)
END SmartHospitalNetworkSimulation
```

5.4 Visual Comparison Across All Scenarios

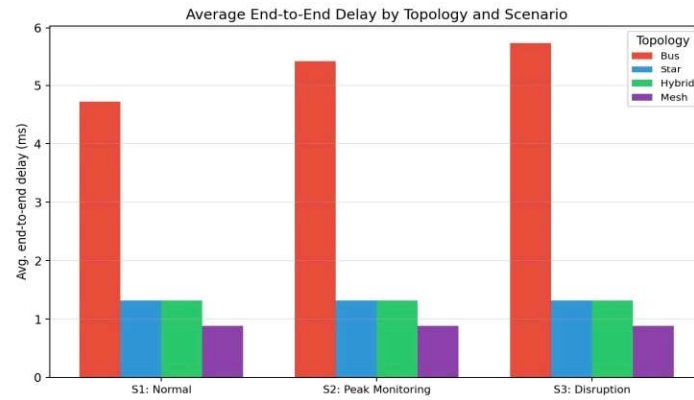


Figure 5.1 — Average end-to-end delay by topology and scenario.

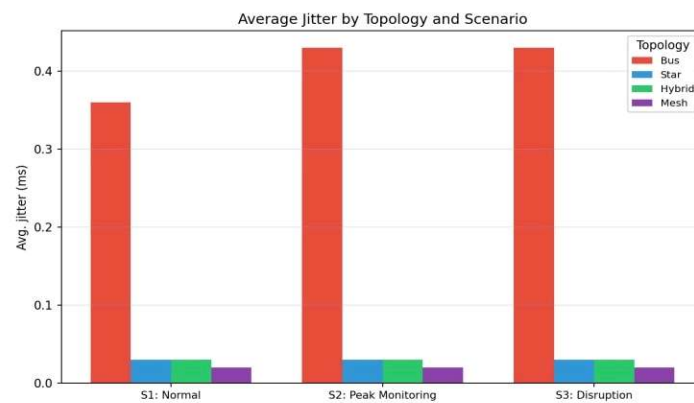


Figure 5.2 — Average jitter by topology and scenario.

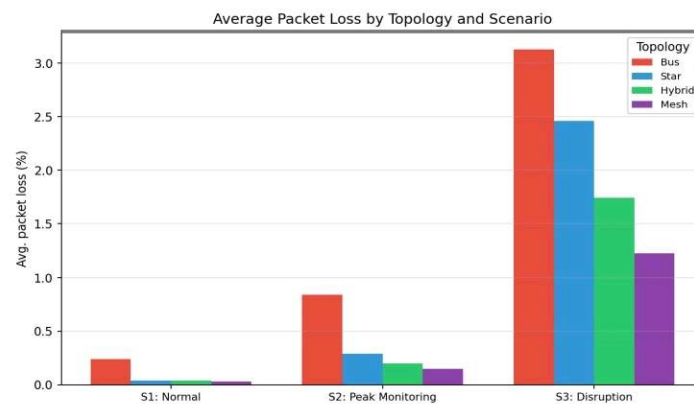


Figure 5.3 — Average packet loss by topology and scenario.

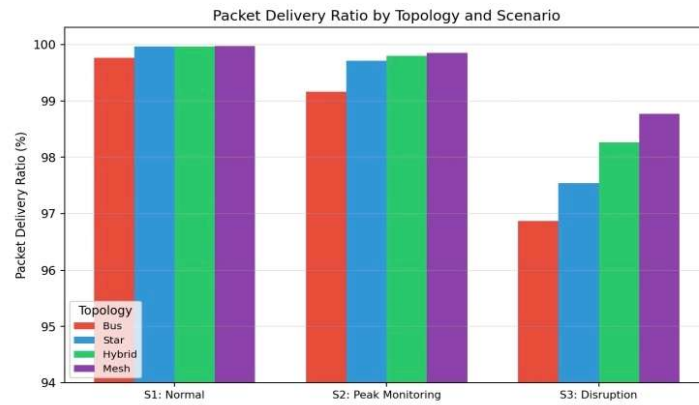


Figure 5.4 — Packet delivery ratio by topology and scenario.

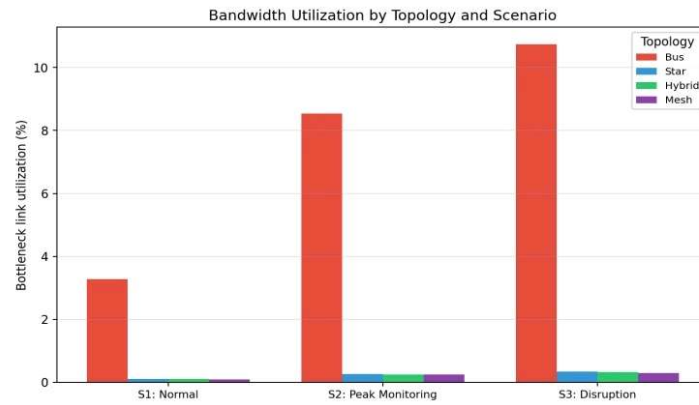


Figure 5.5 — Bottleneck link utilization by topology and scenario.

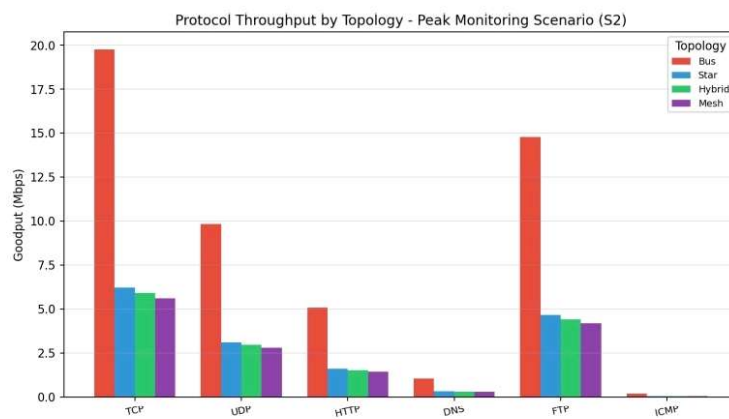


Figure 5.6 — Per-protocol goodput by topology during the peak monitoring scenario (S2).

B.1 Star Topology

Protocol	Scenario	Delay (ms)	Jitter (ms)	Loss (%)	PDR (%)	Throughput (Mbps)
TCP	S1	1.31	0.026	0.034	99.97	2.399
TCP	S2	1.31	0.026	0.164	99.84	6.23
TCP	S3	1.31	0.028	1.404	98.6	4.259
UDP	S1	1.45	0.062	0.041	99.96	1.2
UDP	S2	1.45	0.066	0.42	99.58	3.107
UDP	S3	1.46	0.065	3.518	96.48	2.084
HTTP	S1	1.35	0.037	0.031	99.97	0.614
HTTP	S2	1.34	0.037	0.168	99.83	1.595
HTTP	S3	1.34	0.037	1.407	98.59	1.09
DNS	S1	1.29	0.022	0.045	99.95	0.127
DNS	S2	1.29	0.024	0.418	99.58	0.328
DNS	S3	1.29	0.023	3.518	96.48	0.22
FTP	S1	1.27	0.018	0.027	99.97	1.792
FTP	S2	1.27	0.018	0.168	99.83	4.651
FTP	S3	1.27	0.018	1.409	98.59	3.18
ICMP	S1	1.25	0.014	0.039	99.96	0.023
ICMP	S2	1.25	0.014	0.41	99.59	0.058
ICMP	S3	1.26	0.013	3.51	96.49	0.039

B.2 Bus Topology

Protocol	Scenario	Delay (ms)	Jitter (ms)	Loss (%)	PDR (%)	Throughput (Mbps)
TCP	S1	5.11	0.4	0.345	99.66	7.654
TCP	S2	6.7	0.442	1.085	98.91	19.751
TCP	S3	7.43	0.476	2.551	97.45	13.471
UDP	S1	6	0.648	0.431	99.57	3.823
UDP	S2	7.05	0.868	1.583	98.42	9.826
UDP	S3	7.47	0.8	4.946	95.05	6.57
HTTP	S1	4.77	0.399	0.115	99.89	1.964
HTTP	S2	5.14	0.442	0.397	99.6	5.091
HTTP	S3	5.34	0.488	1.701	98.3	3.479
DNS	S1	4.14	0.244	0.139	99.86	0.405
DNS	S2	4.23	0.275	0.595	99.41	1.048
DNS	S3	4.21	0.289	3.705	96.3	0.703
FTP	S1	4.5	0.287	0.269	99.73	5.719
FTP	S2	5.67	0.371	0.858	99.14	14.782

Protocol	Scenario	Delay (ms)	Jitter (ms)	Loss (%)	PDR (%)	Throughput (Mbps)
FTP	S3	6.17	0.357	2.279	97.72	10.087
ICMP	S1	3.81	0.203	0.146	99.85	0.072
ICMP	S2	3.77	0.175	0.492	99.51	0.187
ICMP	S3	3.79	0.19	3.616	96.38	0.125

B.3 Mesh Topology

Protocol	Scenario	Delay (ms)	Jitter (ms)	Loss (%)	PDR (%)	Throughput (Mbps)
TCP	S1	0.87	0.016	0.03	99.97	2.159
TCP	S2	0.87	0.017	0.087	99.91	5.611
TCP	S3	0.87	0.017	0.705	99.29	3.861
UDP	S1	0.97	0.04	0.04	99.96	1.08
UDP	S2	0.97	0.041	0.206	99.79	2.802
UDP	S3	0.97	0.04	1.751	98.25	1.91
HTTP	S1	0.9	0.022	0.037	99.96	0.553
HTTP	S2	0.9	0.023	0.095	99.91	1.436
HTTP	S3	0.9	0.027	0.701	99.3	0.988
DNS	S1	0.86	0.016	0.03	99.97	0.114
DNS	S2	0.86	0.015	0.199	99.8	0.296
DNS	S3	0.86	0.014	1.751	98.25	0.202
FTP	S1	0.85	0.012	0.034	99.97	1.612
FTP	S2	0.85	0.013	0.089	99.91	4.19
FTP	S3	0.85	0.013	0.708	99.29	2.882
ICMP	S1	0.84	0.009	0.03	99.97	0.02
ICMP	S2	0.84	0.009	0.204	99.8	0.053
ICMP	S3	0.84	0.009	1.758	98.24	0.036

B.4 Hybrid Topology

Protocol	Scenario	Delay (ms)	Jitter (ms)	Loss (%)	PDR (%)	Throughput (Mbps)
TCP	S1	1.31	0.026	0.036	99.96	2.279
TCP	S2	1.31	0.025	0.122	99.88	5.921
TCP	S3	1.31	0.026	0.994	99.01	4.063
UDP	S1	1.45	0.058	0.041	99.96	1.14
UDP	S2	1.45	0.058	0.286	99.71	2.956
UDP	S3	1.45	0.068	2.487	97.51	2.001
HTTP	S1	1.34	0.035	0.035	99.96	0.583
HTTP	S2	1.34	0.037	0.113	99.89	1.516
HTTP	S3	1.35	0.035	0.99	99.01	1.04
DNS	S1	1.29	0.021	0.04	99.96	0.12

Protocol	Scenario	Delay (ms)	Jitter (ms)	Loss (%)	PDR (%)	Throughput (Mbps)
DNS	S2	1.29	0.023	0.282	99.72	0.312
DNS	S3	1.29	0.022	2.476	97.52	0.211
FTP	S1	1.27	0.019	0.031	99.97	1.702
FTP	S2	1.27	0.017	0.123	99.88	4.421
FTP	S3	1.27	0.017	1.001	99	3.034
ICMP	S1	1.25	0.014	0.03	99.97	0.021
ICMP	S2	1.25	0.012	0.29	99.71	0.055
ICMP	S3	1.26	0.014	2.479	97.52	0.038

The screenshot displays the Quetext plagiarism checker interface. The browser address bar shows the URL: `quetext.com/reports/58ae92a7fc5a971590db9`. The page title is "Modern hospitals depend on...". The left sidebar contains navigation options: "New Check", "Reports", "Folders", "Citations", "Bulk File Scan", "AI Summarizer", "AI Humanizer", "Developer Tools", "API", and "Chrome Extension". The main content area shows the text being checked, with a highlighted sentence: "rather than presenting the numbers as more precise than they are." The right sidebar displays the results: "Plagiarism 2%", "Remarks Total: 0", and "AI Score N/A". A message states: "Our detector is highly confident that the text is written by AI. Please use the below arrow keys to see Plagiarism and AI Results". Below this, a "Cite Source" button is visible. The "Plagiarism Matches" section shows a match from "drcro.ai" with "100% similar". The match text is: "... times as effective as Basic" over-reads a single stated ceiling on each side into a precise multiplier that Amazon's own language doesn't actually claim. Neither figure comes with the underlying sample or methodology published alongside it, which is worth acknowledging in an treating the numbers as more precise they are. They're used as a directional signal..."